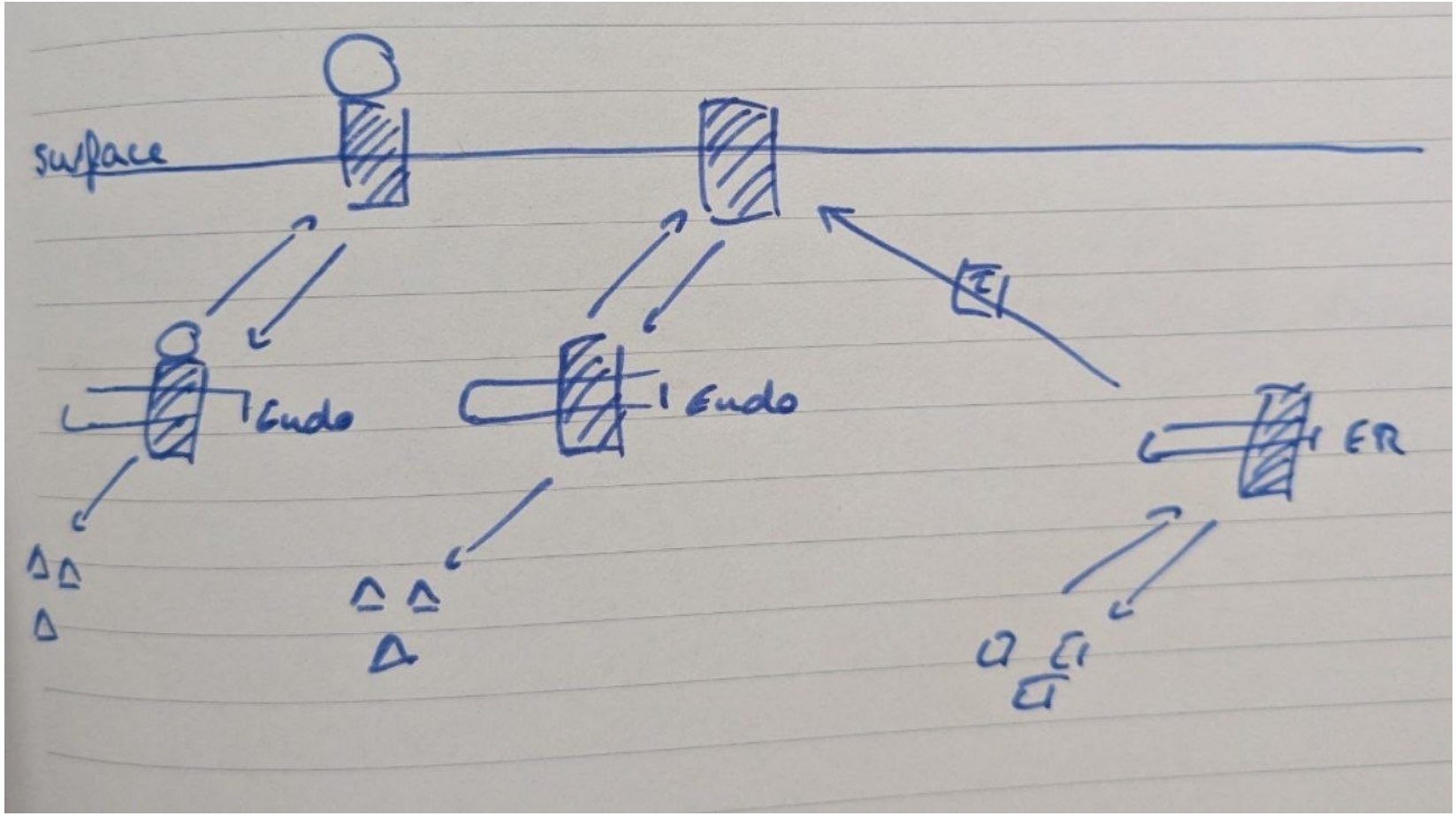
Results TCR-modeling



ODE-equations (reactions)

```
       Educt ->      Product                                                                  Rate
             ->       TCR_ER          k_P / (1 + k_IM_1 * (TCR_Surface + TCRL_Surface * ligand))
      TCR_ER ->                                                                        k_B * TCR_ER
      TCR_ER ->  TCR_Golgi_1 k_E * TCR_ER / (1 + k_IM_3 * (TCR_Surface + TCRL_Surface * ligand))
 TCR_Golgi_1 ->  TCR_Golgi_2                                                     k_E * TCR_Golgi_1
 TCR_Golgi_2 ->  TCR_Surface                                                     k_E * TCR_Golgi_2
 TCR_Surface ->     TCR_Endo                                                     k_I * TCR_Surface
    TCR_Endo ->  TCR_Surface             k_PM_4 * (TCR_Surface + TCRL_Surface*ligand) * TCR_Endo
TCRL_Surface ->    TCRL_Endo                                       k_I_lig * TCRL_Surface * ligand
   TCRL_Endo -> TCRL_Surface      k_PM_4_lig * (TCRL_Surface + TCR_Surface) * TCRL_Endo * ligand
    TCR_Endo ->                                                                     k_D * TCR_Endo
   TCRL_Endo ->                                                         k_D_lig * TCRL_Endo * ligand
```

Observables (map model to data)

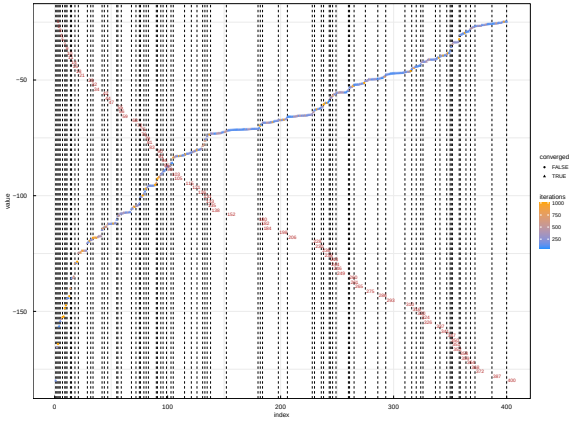
total_TCR = log2(TCR_ER + TCR_Endo + TCRL_Endo + TCR_Golgi_1 + TCR_Golgi_2 + TCR_Surface + TCRL_Surface + offset_Total) + scale_Total

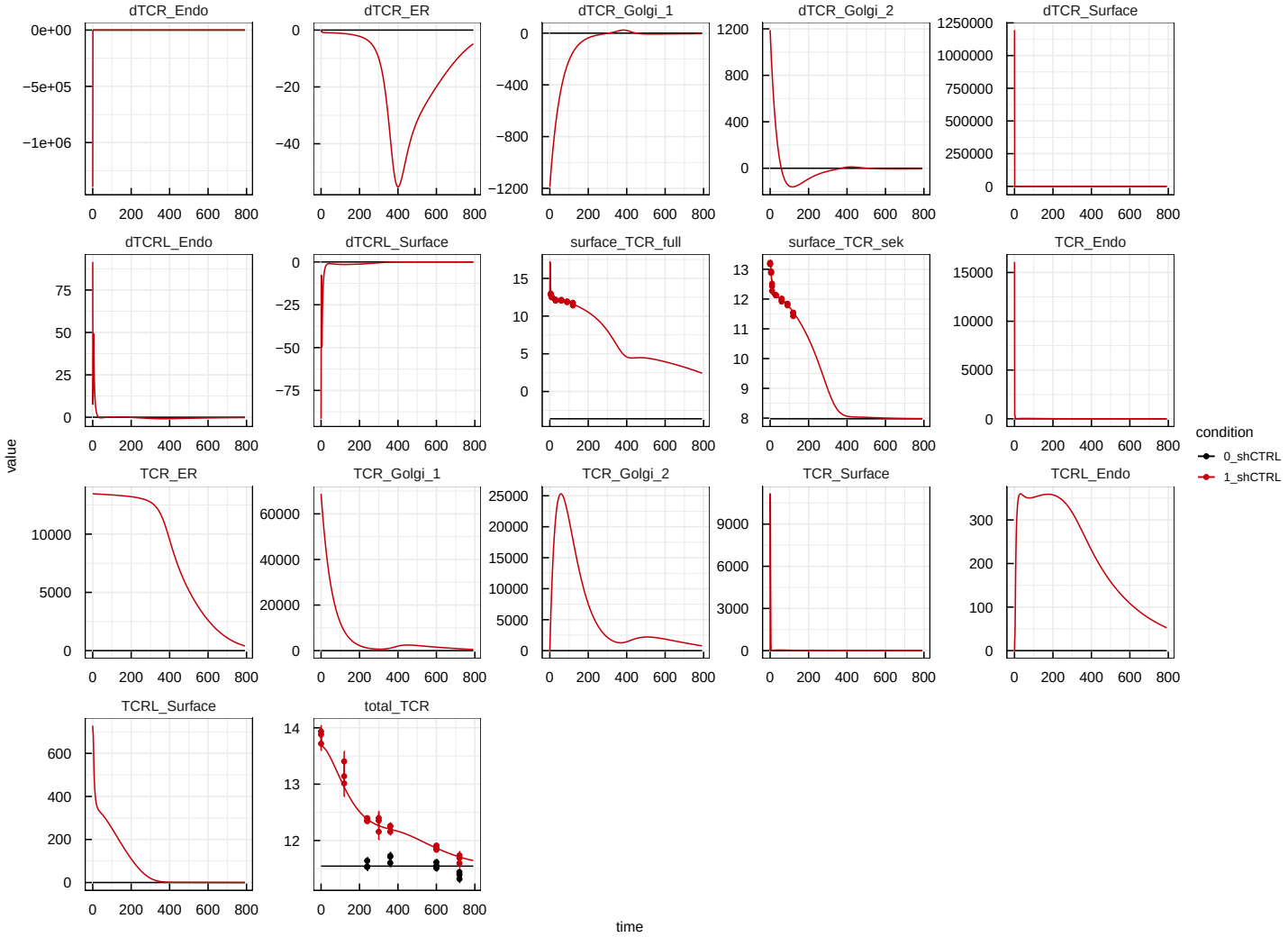
surface_TCR_sek = log2(TCRL_Surface + offset_Surface_sec) + scale_Surface

surface_TCR_full = log2(TCRL_Surface + TCR_Surface + offset_Surface_full) + scale_Surface

Additionally: Add steady-state constraint for data without ligand, use that the derivative has to be 0

Results


value
index
converged
FALSE
TRUE
parations


dTCR_Endo
dTCR_ER
dTCR_Golgi_1
dTCR_Golgi_2
dTCR_Surface
dTCRL_Endo
dTCRL_Surface
surface_TCR_full
surface_TCR_sek
TCR_Endo
TCR_ER
TCR_Golgi_1
TCR_Golgi_2
TCR_Surface
TCRL_Endo
TCRL_Surface
total_TCR
value
time
condition
0_shCTRL
1_shCTRL

What’s the problem?

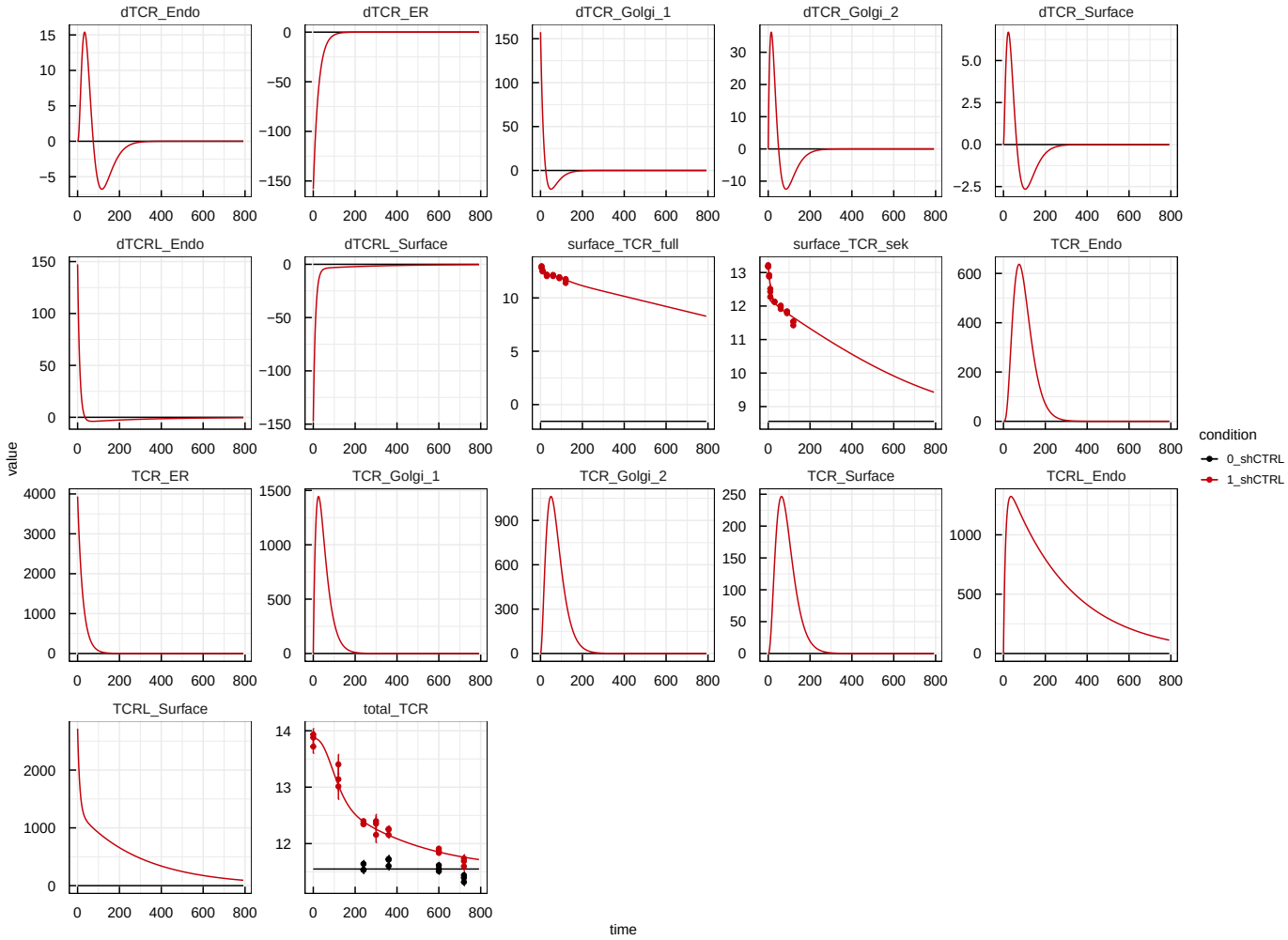
- Model too big for amount of data
- Overlap in data rather small → oszillation possible
- Too many feedbacks
- No profile-based model reduction possible
- No estimation of parameters possible

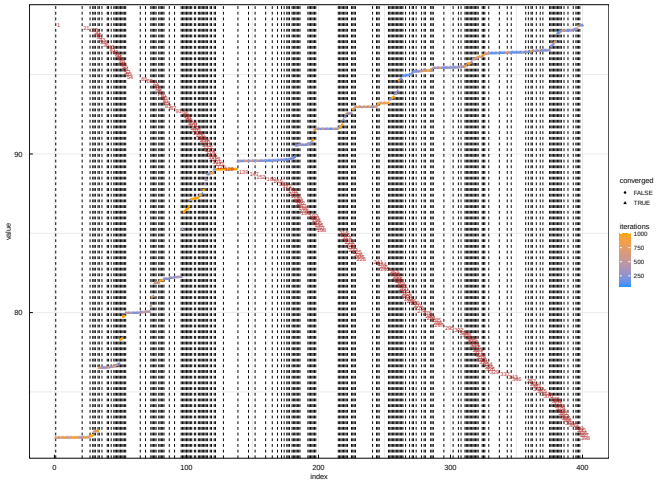
Trial and Error: Remove feedbacks

- Set k_IM_3 to 0
- Remove activation by TCR_Surface in

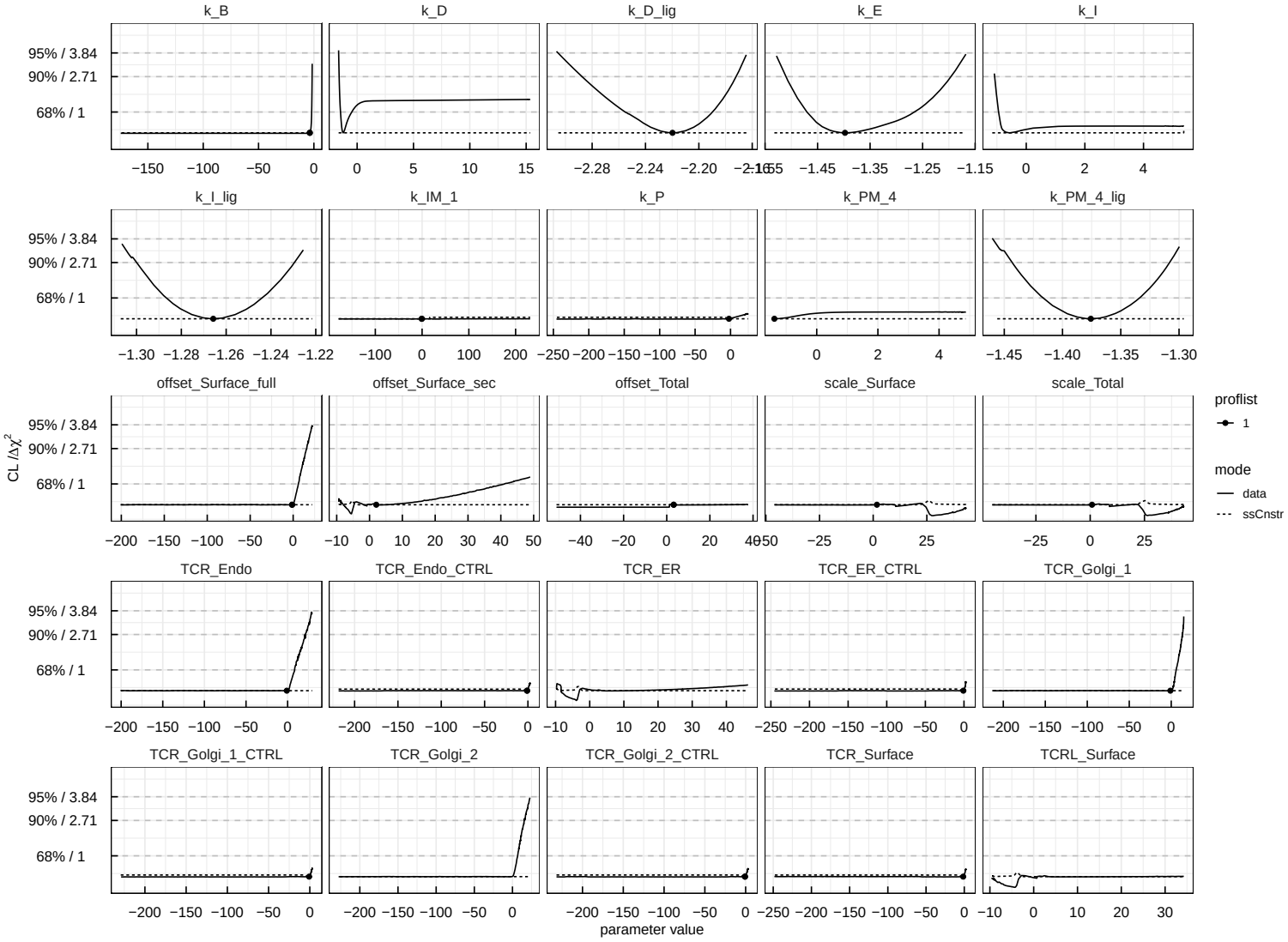
  TCR_Endo → TCR_Surface reaction

Results


dTCR_Endo
dTCR_ER
dTCR_Golgi_1
dTCR_Golgi_2
dTCR_Surface
dTCRL_Endo
dTCRL_Surface
surface_TCR_full
surface_TCR_sek
TCR_Endo
TCR_ER
TCR_Golgi_1
TCR_Golgi_2
TCR_Surface
TCRL_Endo
TCRL_Surface
total_TCR
value
time
condition
0_shCTRL
1_shCTRL


value
index
converged
FALSE
TRUE
iterations
1000
750
500
250

First Profiles



Next steps, questions and ideas

- Look at profiles paths and possible reduction
- Can we measure the steady state fot the surface?
- Maybe more overlap in time-points
- Why is the ligand washed out?